



Welcome to INGB417 Facilities Design 2026

Lecturer: Mrs Willemien Kruger

Contact nr: 083 7112246

Email: Willemien.Kruger@nwu.ac.za

Office: 109

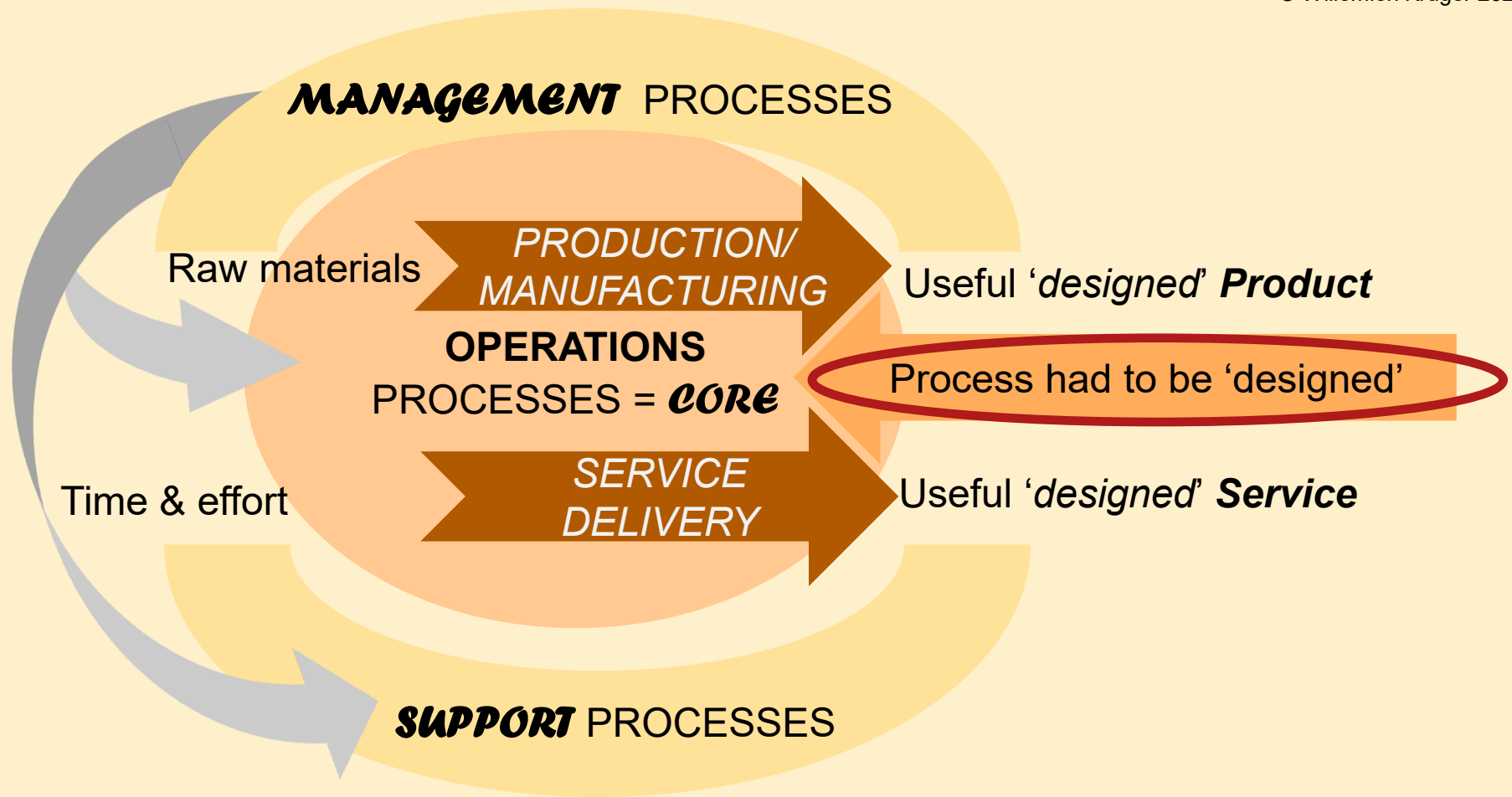
Consulting hours : any day
except Thursdays



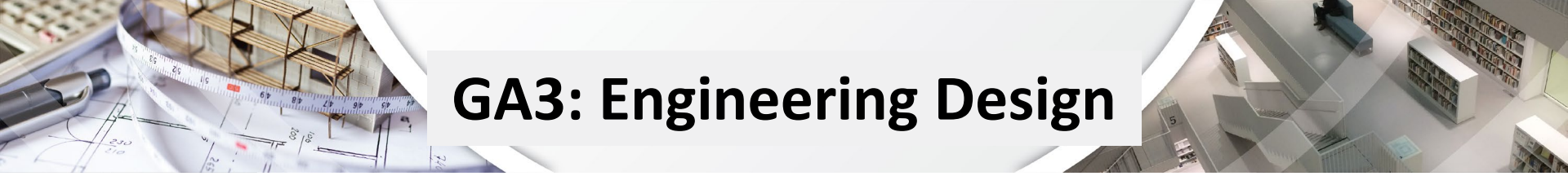
BUSINESS 'PROCESSES'



© Willemien Kruger 2024



Financial support, Administration, Purchasing, Inventory management, Marketing, Sales & Distribution, Research & Development, Projects/Programs, Facilities and equipment, Information technology/Systems



GA3: Engineering Design

The purpose of this module is to develop your ability to
DESIGN creative solutions,
for complex engineering problems and
DESIGN

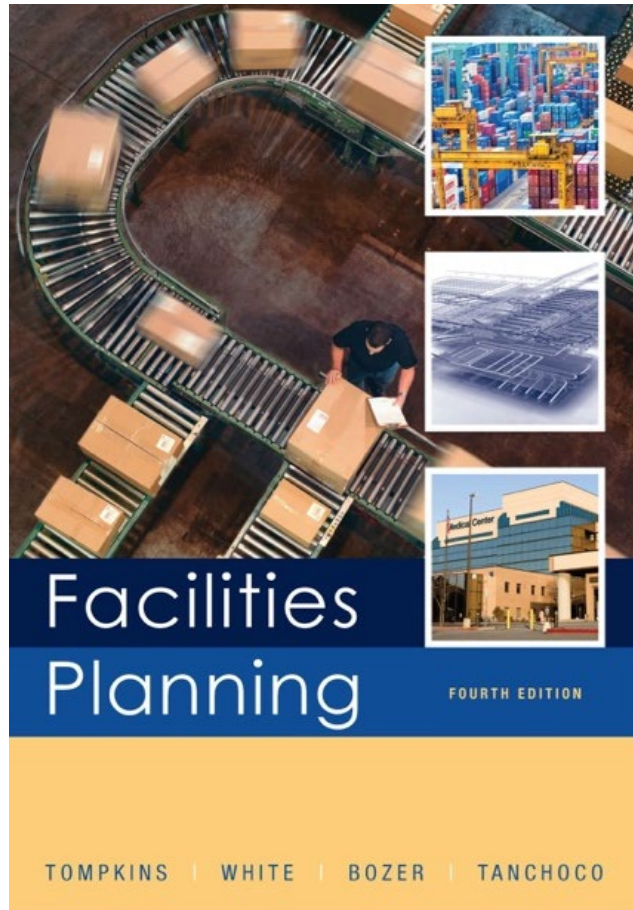
systems, components or processes to
meet identified needs(=GA3)

In the context of doing a FACILITY DESIGN....

**“CREATING PLACES THAT
MAXIMIZE EFFICIENCY”**

**INGB417 includes both FACILITY DESIGN and
PRODUCTION PROCESS DESIGN
(they affect each other)**

Textbooks

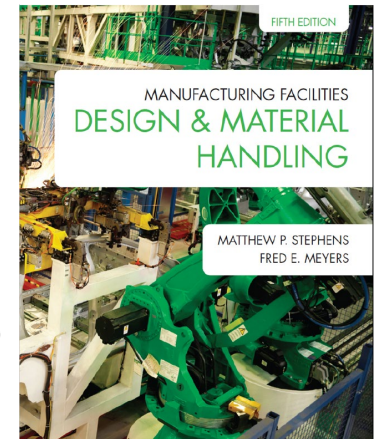


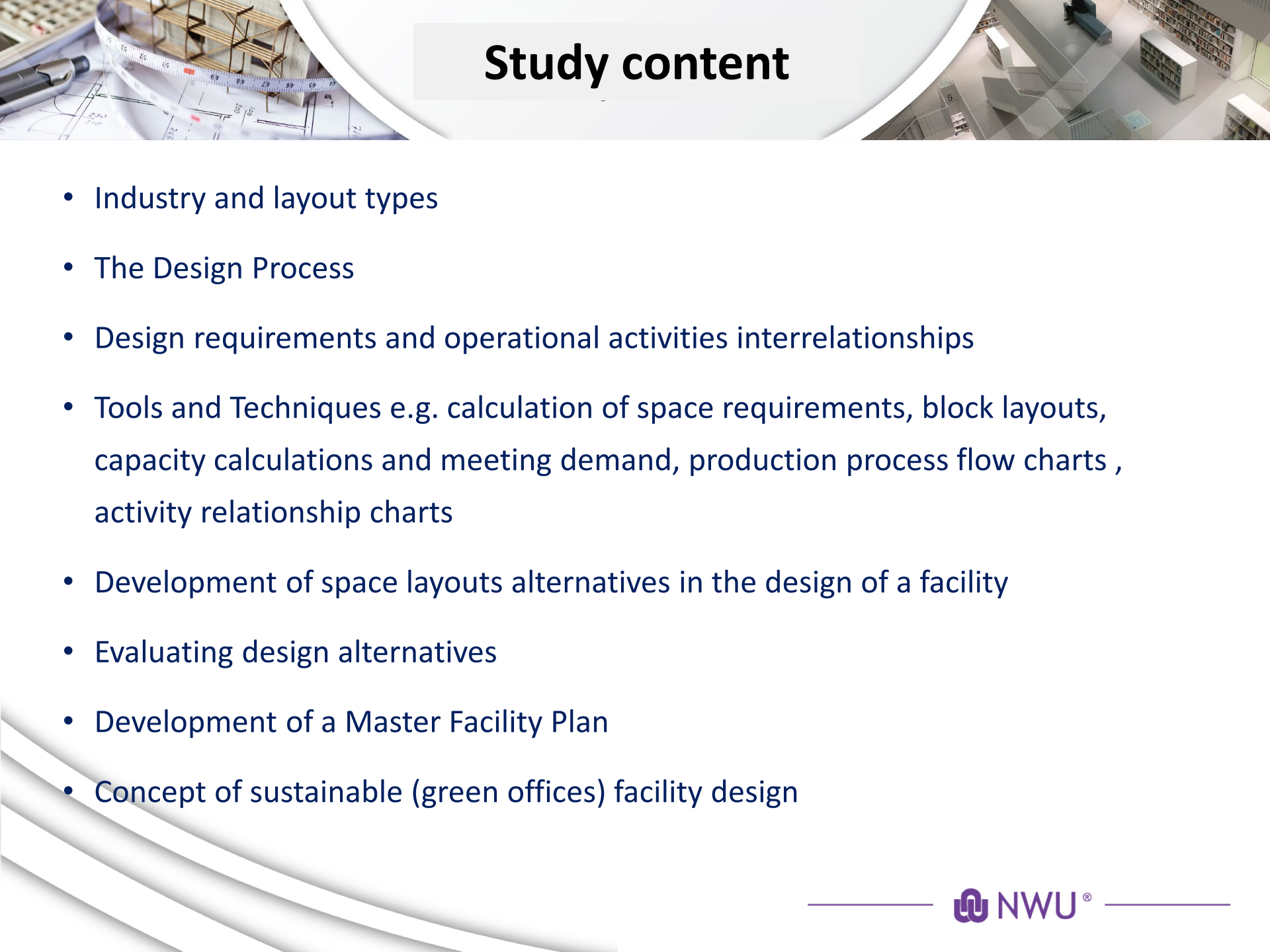
Prescribed textbook:

Tompkins JA, White JA, Bozer YA, Tanchoco JMA, *Facilities Planning*, 4th Edition, Wiley & Sons, 2010, ISBN : 978-0-470-44404-7.

Additional textbook:

Stephens, Matthew P, *Manufacturing Facilities Design & Material Handling*, 5th or 6th edition, Purdue University Press, 2019, ISBN: 978-1-61249-573-6





Study content

- Industry and layout types
- The Design Process
- Design requirements and operational activities interrelationships
- Tools and Techniques e.g. calculation of space requirements, block layouts, capacity calculations and meeting demand, production process flow charts , activity relationship charts
- Development of space layouts alternatives in the design of a facility
- Evaluating design alternatives
- Development of a Master Facility Plan
- Concept of sustainable (green offices) facility design

Assessment Plan

Participation mark (50% ratio)
(Need 40% to participate in exam)

Exam/
GA Assignment
(50% ratio)

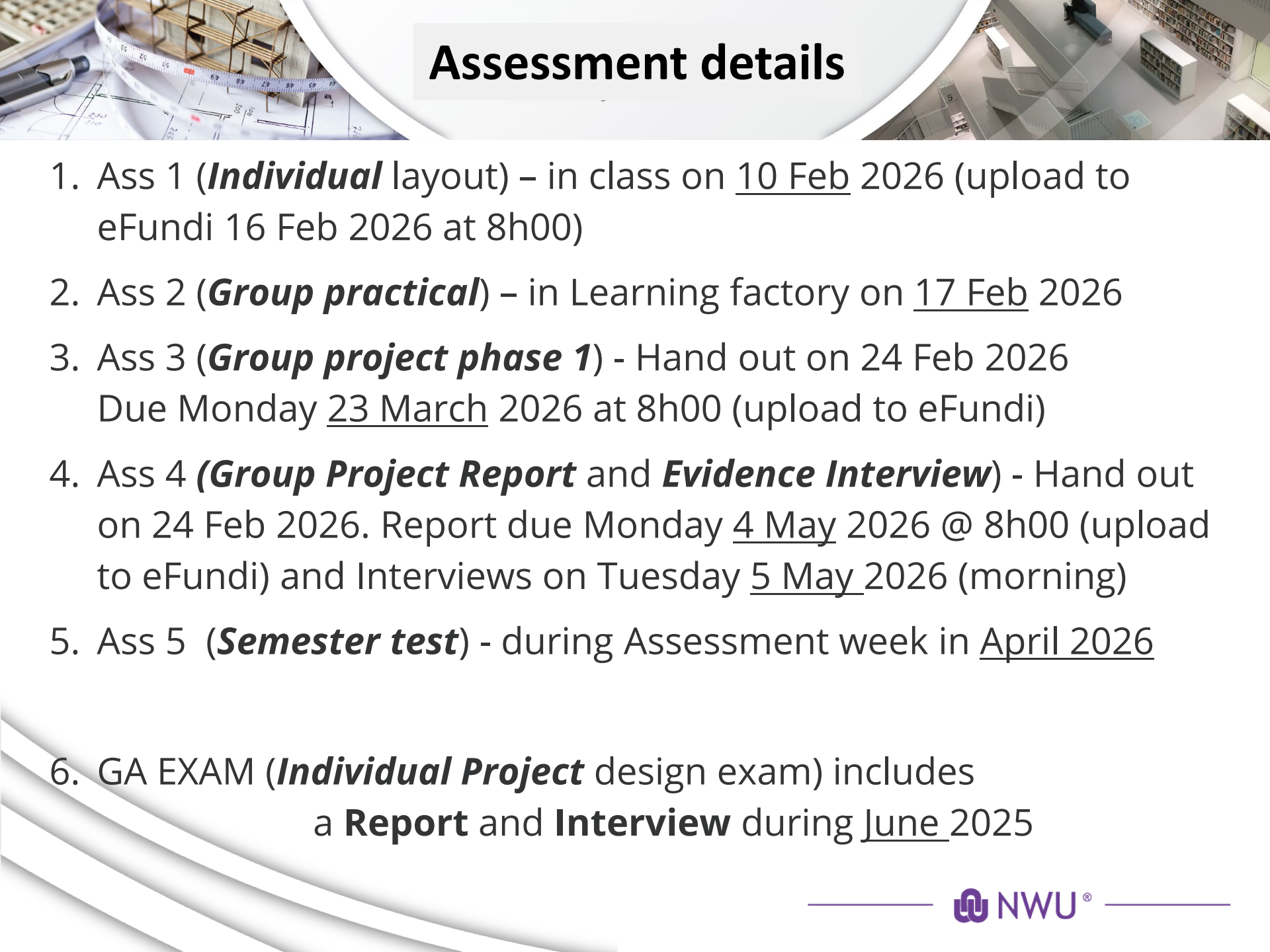
1 Assignment in class = 10%
1 Practical = 10%
1 Group Project (2 parts) = **60%**
1 Semester Test = 20%

Subminimum of 40%
Pass GA3 (YES/NO)

My expectation is that you ATTEND CLASSES – this is a **contact based** university, and this module is especially NB, some classes are practical work and I say a lot of things in class that may be missed on the slides. So you will NOT have enough context with only slides.

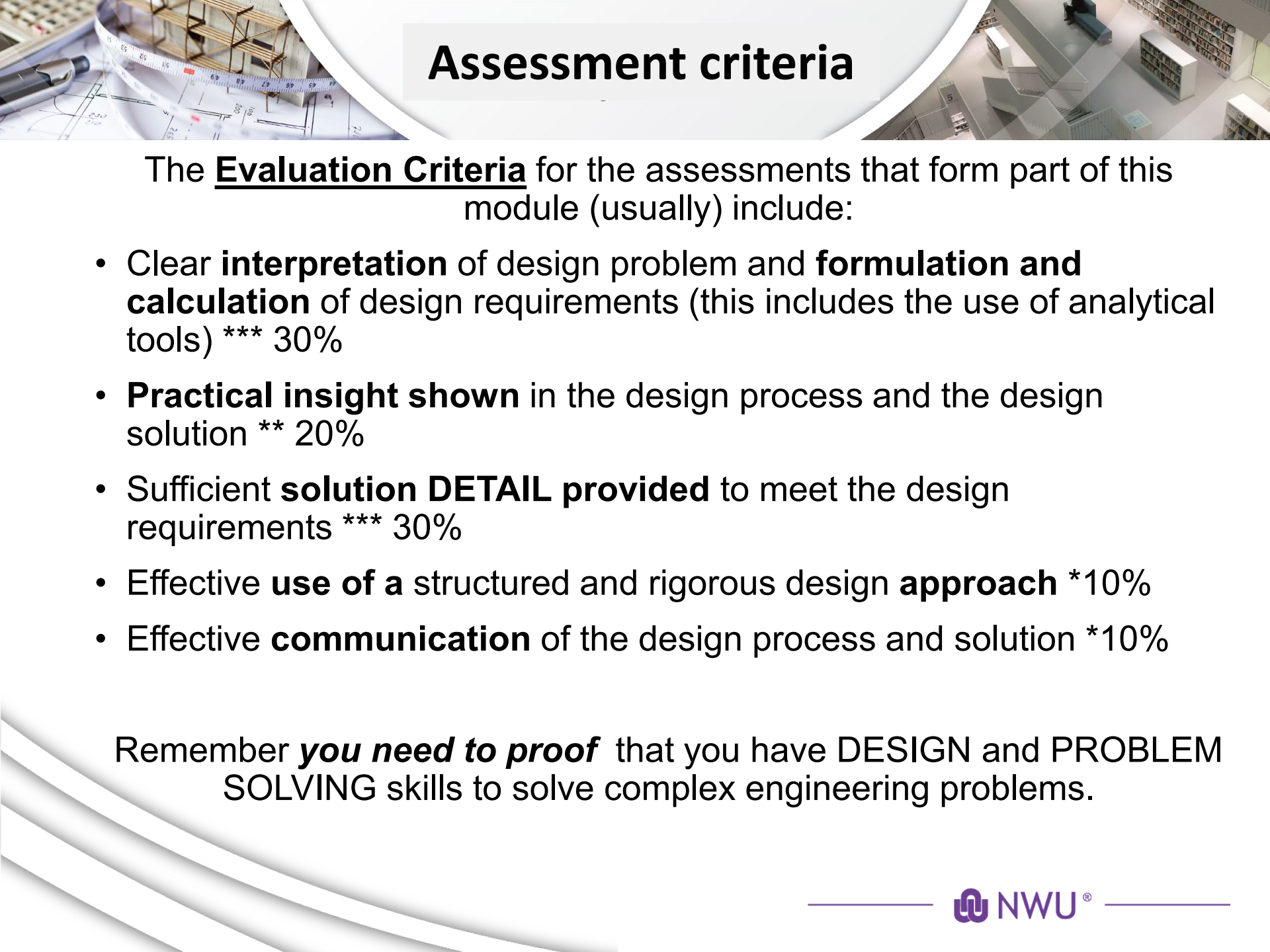
There are only 10 classes and 1 outing...so I NEED TO SEE YOU in class!

This is a BIG module, with a BIG assignment as the main focus...



Assessment details

1. Ass 1 (**Individual** layout) – in class on 10 Feb 2026 (upload to eFundi 16 Feb 2026 at 8h00)
2. Ass 2 (**Group practical**) – in Learning factory on 17 Feb 2026
3. Ass 3 (**Group project phase 1**) - Hand out on 24 Feb 2026
Due Monday 23 March 2026 at 8h00 (upload to eFundi)
4. Ass 4 (**Group Project Report** and **Evidence Interview**) - Hand out on 24 Feb 2026. Report due Monday 4 May 2026 @ 8h00 (upload to eFundi) and Interviews on Tuesday 5 May 2026 (morning)
5. Ass 5 (**Semester test**) - during Assessment week in April 2026
6. GA EXAM (**Individual Project** design exam) includes a **Report** and **Interview** during June 2025



Assessment criteria

The **Evaluation Criteria** for the assessments that form part of this module (usually) include:

- Clear **interpretation** of design problem and **formulation and calculation** of design requirements (this includes the use of analytical tools) *** 30%
- **Practical insight shown** in the design process and the design solution ** 20%
- Sufficient **solution DETAIL provided** to meet the design requirements *** 30%
- Effective **use of a** structured and rigorous design **approach** *10%
- Effective **communication** of the design process and solution *10%

Remember ***you need to proof*** that you have DESIGN and PROBLEM SOLVING skills to solve complex engineering problems.

Exam Rubric

If you hand in
an Incomplete
or NO Design
or Alternatives
or an
Incomplete or
No MFP =
FAIL!!!

GA 3: Engineering Design

Design creative solutions for complex engineering problems and design *systems, components* or *processes* to meet identified needs.

EVALUATION CRITERIA

	Weighting	Mark (Interview)	Mark (Report)
Clear interpretation of the design problem and <i>formulation/calculation of design requirements or specifications</i> . (includes appropriate use of <i>analytical tools</i> and techniques.)	/30		
Practical <i>insight shown</i> during the design process and in the design solution	/20		
Sufficient <i>detail provided in the solution</i> to meet the design requirements (Receiving & Dispatch, Production processes, Storage, Warehouses, Offices, Support functions all in MFP)	/30		
Effective use of a structured and rigorous <i>design process</i> (Report structure corresponding to process steps)	/10		
Effective <i>communication</i> of the design process and solution	/10		



How the module will practically work

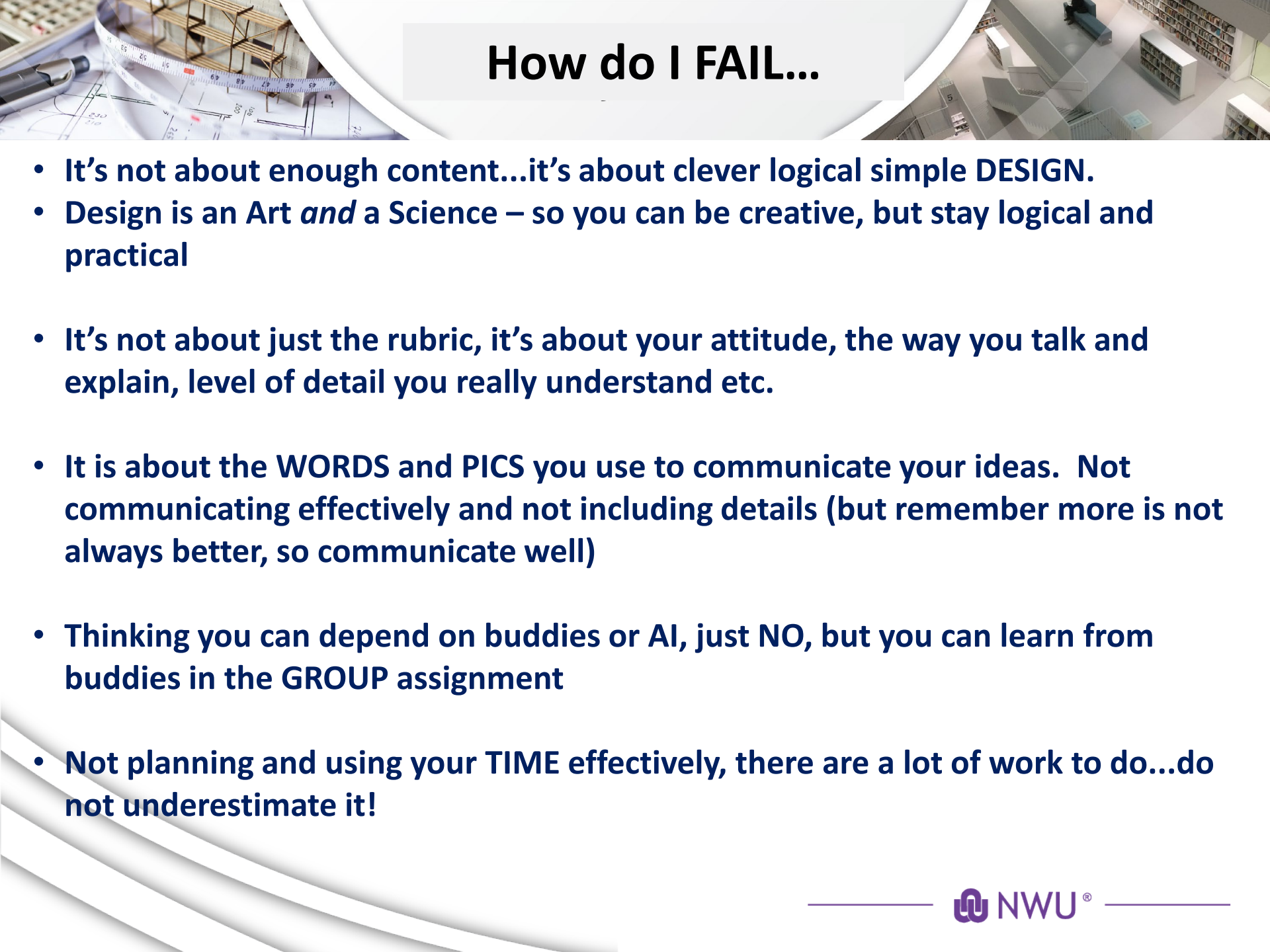


- **ALL classes are presented in person – attend the classes!**
- **Read the textbooks (this is Self Study)**
- **Do the assessments to the best of your ability (DO NOT COPY and PASTE, seek shortcuts with or without AI, or compare too much to your peers)**
- **You need to learn the DESIGN skill yourself**
- **Be creative, think outside of the box (this is an exciting module!)**
- **Note: AI is really bad with facilities design, so do not even try.**
- **This is an opportunity to learn a really useful skill, use it**

Lecture plan 2026

INGB417 FACILITIES DESIGN MODULE PLAN

contact session	day	date	time planned	Plan	study unit	Textbook (Tompkins)	Textbook (Stephens)
1		10-Feb	8h00-10h45	417Overview, Industry types, Layout types lectures and Ass1 Practical measurement	1		
			12h45-14h15	<i>practical help with draw.io or other software for Ass1&2</i>			
2		17-Feb	8h00-10h45	Learning Factory Practical 3hrs			
			12h45-14h15	Assessments feedback			
3		24-Feb	8h00-10h45	Design process lecture +understanding GA3 + Assignment3&4 groups assign, handout and explain	2	Ch1	Ch1
			12h45-14h15	Assessment 3&4 group work opportunity + questions + recap capacity calculations			
4		03-Mar	8h00-10h45	Production process Design and Requirements (Demand to Capacity) calculations [demi]	2,3,4	Ch2	Ch2,3,4,5
			12h45-14h15	no class			
5		10-Mar	8h00-10h45	Systematic Layout Planning (SLP) lecture + Tutorial work	5,6	Ch3,6	Ch 6,7,13
			12h45-14h15	Flow, activities and Space requirements lecture			
6		17-Mar	8h00-10h45	Material handling theory and variety	6	Ch5	Ch10,11
			12h45-14h15	if needed extra tutorial on SLP			
7		24-Mar	8h00-10h45	Personnel offices, auxiliary services, warehouse and production stations	7	Ch4,7,9	Ch8,9,12
			12h45-14h15	no class			
Assessment5=SemTest 30March -10 April 2026							
8		14-Apr	8h00-10h45	Alternatives and Evaluation and MFP incl Business case and FEEDBACK on Ass3	8	Ch11,12	Ch14, 16
			12h45-14h15	available for questions regarding Ass4. <i>Show good & bad reports/MFP's.</i>			
		21-Apr	No class	Work on GROUP PROJECT. Redo Ass3 based on feedback and do Ass4 DESIGN			
		04-May Monday		at 8h00 Ass4 report due date on eFundi			
9		05-May	8h00-10h45	Group project interviews			
			12h45-14h15	Discuss Ass4 feedback as prep for exam			
10		12-May	8h00-10h45	Servicescape and SMART offices			
			12h45-14h15	OUTING ?			
		19-May	9h00-11h00	OUTING ?			



How do I FAIL...

- It's not about enough content...it's about clever logical simple DESIGN.
- Design is an Art *and* a Science – so you can be creative, but stay logical and practical
- It's not about just the rubric, it's about your attitude, the way you talk and explain, level of detail you really understand etc.
- It is about the WORDS and PICS you use to communicate your ideas. Not communicating effectively and not including details (but remember more is not always better, so communicate well)
- Thinking you can depend on buddies or AI, just NO, but you can learn from buddies in the GROUP assignment
- Not planning and using your TIME effectively, there are a lot of work to do...do not underestimate it!



AI struggle with this

Here is the updated image with welding machines and human welders instead of computers. Let me know if you need any further refinements!

you don't know how welding stations look like do you?

You're right to call that out! Welding stations have specific elements like welding booths, fume extractors, protective barriers, welding torches, sparks, and workers wearing PPE (helmets, gloves, and protective aprons). Let me refine the image to reflect a more realistic welding facility.

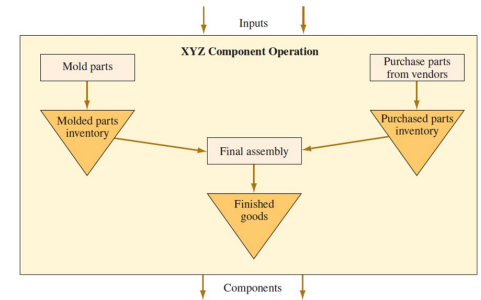


Here is the updated image with a more realistic welding facility, including proper welding stations, protective barriers, ventilation, and workers in full PPE. Let me know if this better captures what you're looking for!

You need to's...

- You need to be able to understand production processes....

Raw material is what? Where?



Finished good is what?

- You need to be able to visualize the **flow of materials** and how it changes through the production process
- You need to understand the concept of a '**workstation**' (Machine, operator, WIP, scrap)
- and the idea of 'functional areas' and 'department' groups
- You need to understand movement between groups
- You need to understand scale...making 1000 vs 10 000
- – group together
- You need to think logically!
- You need to understand floor spaces and open spaces
- You need to think about the inside of a building but also the outside logistics, how to get there, where to park and the flow of people logic

Examples –

- There are GOOD, OK and BAD facility designs
- It can be plain wrong, but if you can explain the logic and assumptions behind your design, show it clearly and have evidence
- Show physical size of an MFP and a report
- Show other slideshow with samples

Are you looking forward to learning the skill of DESIGN??